Mathematics for Machine Learning

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Equation of a Line & Vectors

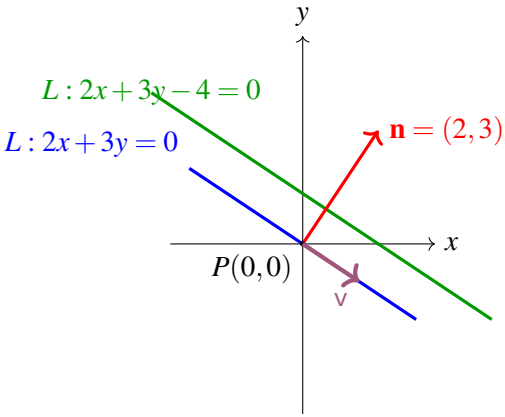
The equation of a line in 2D:

$$ax + by + c = 0$$

A normal vector to the line is $\mathbf{n} = (a, b)$.

A vector has magnitude and direction; basis of $\mathbb{R}^2$ shown.

- Line normal: $\mathbf{n} \cdot (x, y) + c = 0$.
- If $\mathbf{v}$ is parallel to the line, $\mathbf{n} \cdot \mathbf{v} = 0$.



Functions: Domain, Codomain, Range

- A function $f : A \to B$ maps each element of A to exactly one element of B.
- If $A, B \subseteq \mathbb{R}$, then f is called a **real function**.
- If only $B \subseteq \mathbb{R}$ (codomain real), then f is called a **real-valued function**.

$$\begin{tikzcd} A \arrow[r, bend left, "f"] & B \arrow[l, bend left, "f^{-}1"] \end{tikzcd}$$

domain A, codomain B, range $f(A)$

Limit of a Function

Definition: The limit of $f(x)$ as $x \to a$ is the value approached by $f(x)$ when x approaches a.

We require left and right limits to agree:

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L.$$

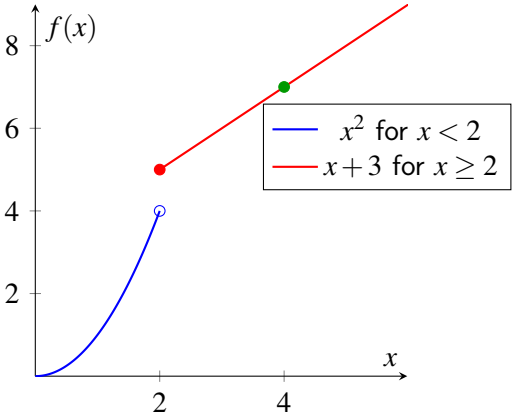
Piecewise example:

$$f(x) = \begin{cases} x & x < 2 \\ x + 3 & \text{otherwise} \end{cases}$$

Left & Right Neighbors near $x = 4$

h	x_L	$f(x_L)$	x_R	$f(x_R)$
1	3.0	6.0	5.0	8.0
0.5	3.5	6.5	4.5	7.5
0.1	3.9	6.9	4.1	7.1
0.01	3.99	6.99	4.01	7.01
0.001	3.999	6.999	4.001	7.001
0.0001	3.9999	6.9999	4.0001	7.0001

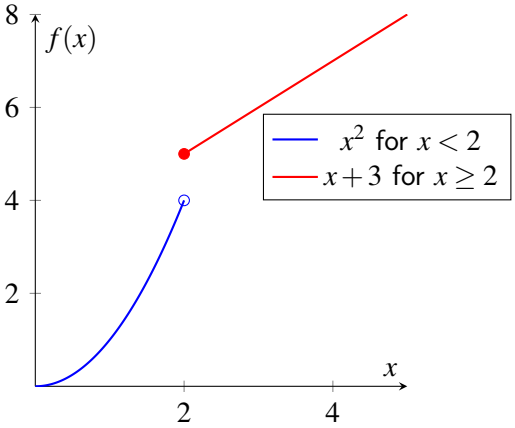
Table: Left and right neighbors of $x = 4$ and corresponding $f(x)$ values



Left & Right Neighbors near $x = 2$

h	x_L	$f(x_L)$	x_R	$f(x_R)$
1	1.0	1.0	3.0	6.0
0.5	1.5	2.25	2.5	5.5
0.1	1.9	3.61	2.1	5.1
0.01	1.99	3.96	2.01	5.01
0.001	1.999	3.996	2.001	5.001
0.0001	1.9999	3.9996	2.0001	5.0001

Table: Left and right neighbors of $x = 2$ and corresponding $f(x)$ values



Continuity and Derivative — Intuition

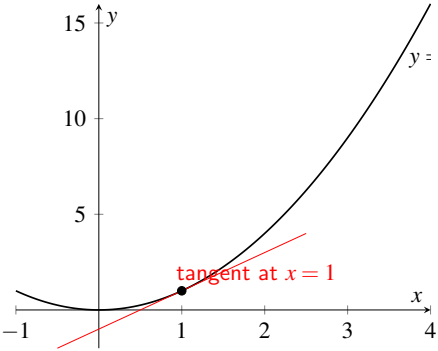
A function is continuous at a if

$$\lim_{x \to a} f(x) = f(a).$$

Derivative as slope of secant lines limit:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

Example: If $f(x) = 2x + 1$, then $f'(x) = 2$.



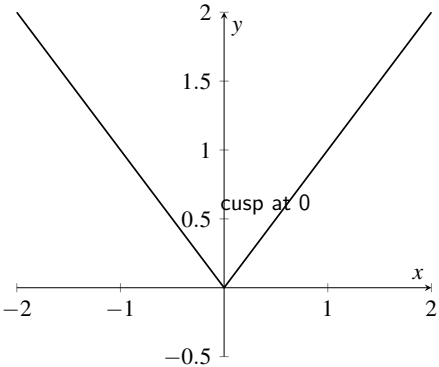
Derivative Examples and Non-differentiability

For $f(x) = x^2$, $f'(x) = 2x$.

For $f(x) = |x|$,

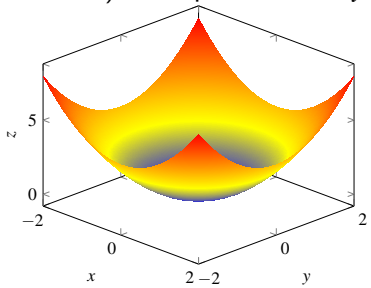
$$f'(x) = \begin{cases} -1 & x < 0 \\ 1 & x > 0 \end{cases}$$

So $f'(0)$ does not exist (left and right derivatives differ).



Multivariate Functions

Bivariate Function $z = f(x,y)$ (two variables) Example: $z = x^2 + y^2$



Trivariate Function $w = f(x,y,z)$ (three variables) Example: $w = x + y + z$

Visualization

Hard to plot 4D, but can use slices or color maps.

n**-variate Function** $f = f(x_1, x_2, \ldots, x_n)$ Example: $f = \sum_{i=1}^{n} x_i$

Visualization

Abstract high-dimensional space.

Partial Derivatives — Example with Two Variables

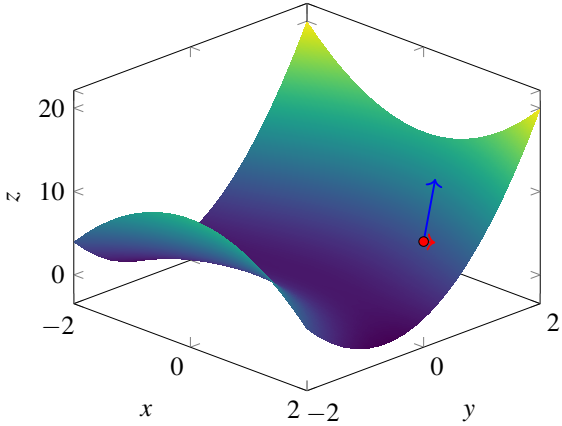
Definition: If $z = f(x,y)$, the partial derivatives measure how z changes with respect to one variable while holding the other constant:

$$\frac{\partial z}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}$$

$$\frac{\partial z}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y+\Delta y) - f(x,y)}{\Delta y}$$

Example: Let $z = x^2 y + 3y^2$

Then $\frac{\partial z}{\partial x} = 2xy \qquad \frac{\partial z}{\partial y} = x^2 + 6y$



Interpretation

$\frac{\partial z}{\partial x}$: slope in x-direction (keep y fixed)

$\frac{\partial z}{\partial y}$: slope in y-direction (keep x fixed).

Partial Derivative Example

Function:

$$z = f(x,y) = 2x + 3y$$

$$\frac{\partial z}{\partial x} = 2, \quad \frac{\partial z}{\partial y} = 3$$

$$f(1,0) = 2 \quad \Rightarrow \quad f(3,4) = 18$$

$$\Delta f = 18 - 2 = 16, \quad \Delta x = 2, \quad \Delta y = 4$$

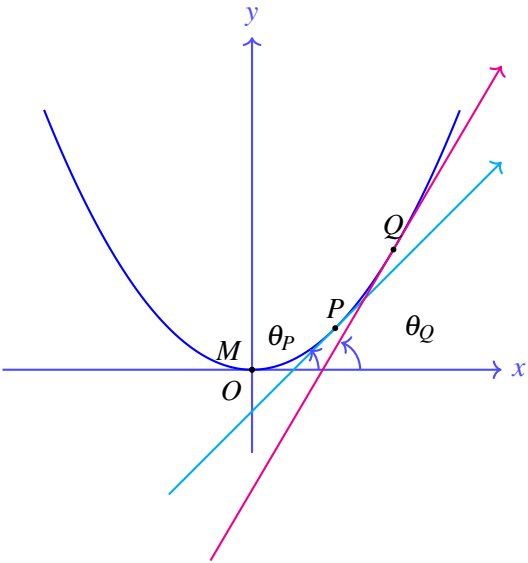
$$\Delta f = \Delta x \cdot 2 + \Delta y \cdot 3 = 16$$

$$\Delta f = \Delta x \frac{\partial z}{\partial x} + \Delta y \frac{\partial z}{\partial y} = 16$$

f	$y=0$	1	2	3	4
$x=0$	0	3	6	9	12
$x=1$	**2**	5	8	11	14
$x=2$	4	7	10	13	16
$x=3$	6	9	12	15	**18**
$x=4$	8	11	14	17	20

$f(x,y) = 2x + 3y$

Derivatives and Optimization



We know: Here

$\theta_P < \theta_Q \Leftrightarrow \tan(\theta_P) < \tan(\theta_Q)$; we know,

$$\left.\frac{dy}{dx}\right|_P = \tan\theta_P, \quad \left.\frac{dy}{dx}\right|_Q = \tan\theta_Q.$$

Angle of tangent at M?

$$\theta_M = 0 \Rightarrow \tan\theta_M = 0 \Rightarrow \frac{dy}{dx} = 0 \text{ at M.}$$

Derivative at a point $x_c = 0 \Rightarrow x_c$ is an extremum point (maximum or minimum), then it can be used for ***optimization****.*

Developing Algorithms: Gradient Methods

Gradient Descent in $\mathbb{R}^n$

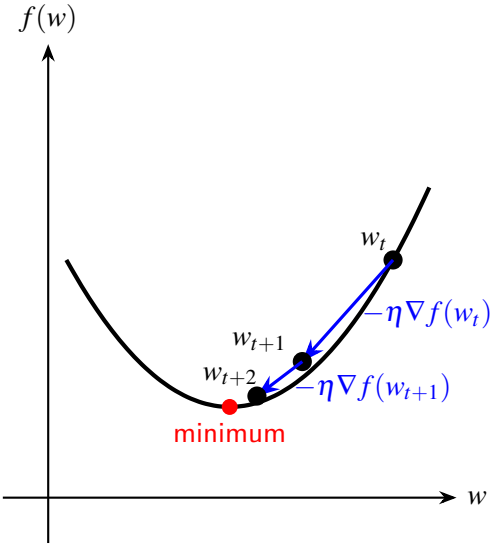
$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta_t \nabla f(\mathbf{w}_t), \quad \mathbf{w}_t \in \mathbb{R}^n$$

with

$$\nabla f(\mathbf{w}) = \begin{bmatrix} \frac{\partial f}{\partial w_1} \\ \frac{\partial f}{\partial w_2} \\ \vdots \\ \frac{\partial f}{\partial w_n} \end{bmatrix} \in \mathbb{R}^n.$$

- **Vectorization:** gradients for batches computed via matrix multiplications in $\mathbb{R}^n$.
- **Curvature:** Hessian $H = \nabla^2 f \in \mathbb{R}^{n \times n}$ guides Newton / Quasi-Newton methods.
- **Deep learning:** backprop applies the chain rule efficiently in matrix form over $\mathbb{R}^n$.

Gradient Descent on a Quadratic Function



What is Gradient Descent?

Goal: Minimize a function $f(x)$

Key Idea: Use local slope information to iteratively move toward lower values

Update Rule:

$$x_{t+1} = x_t - \eta f'(x_t) \tag{1}$$

Interpretation:

- $f'(x_t)$ gives direction of steepest increase
- $-f'(x_t)$ gives direction of steepest decrease
- η controls how far we move (step size)

Extension (Multivariable Case):

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \nabla f(\mathbf{x}_t) \tag{2}$$

Step 1: From Definition of Derivative

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

For small Δx:

$$f'(x) \approx \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

Rearranging:

$$f(x+\Delta x) - f(x) \approx f'(x)\,\Delta x$$

Interpretation:

- Left side: actual change in function
- Right side: approximate change using slope

Step 2: Condition for Moving Toward Minimum

To move toward minimum:

$$f(x+\Delta x) < f(x)$$

$$\Rightarrow f(x+\Delta x) - f(x) < 0$$

Using approximation:

$$f'(x)\Delta x < 0$$

Key Insight:

- The product must be negative
- So Δx must have opposite sign of $f'(x)$

Step 3: Understanding Both Cases

Case 1: $f'(x) > 0$ **(slope positive)**

- Function increasing
- To decrease function: move left
- So $\Delta x < 0$

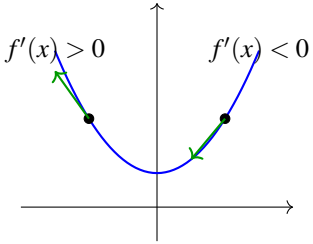
Case 2: $f'(x) < 0$ **(slope negative)**

- Function decreasing
- To decrease further: move right
- So $\Delta x > 0$

Conclusion:

$$\Delta x = -f'(x)$$

Visual Intuition



Observation: Always move in direction $-f'(x)$ to decrease $f(x)$

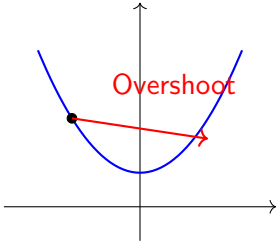
Step 4: What Goes Wrong?

Using:

$$x_{\text{new}} = x - f'(x)$$

Problem:

- Step size may be too large
- Can overshoot minimum
- May oscillate or diverge



Step 5: Remedy (Learning Rate)

Introduce scaling factor η

$$\Delta x = -\eta f'(x), \quad \eta > 0$$

$$x_{\text{new}} = x - \eta f'(x)$$

Effect:

- Controls step size
- Prevents overshooting
- Ensures smooth convergence

Step 6: Generalization to Multivariable Case

$$f(\mathbf{x}+\Delta\mathbf{x}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^T \Delta\mathbf{x}$$

To decrease:

$$\nabla f(\mathbf{x})^T \Delta\mathbf{x} < 0$$

Choose:

$$\Delta\mathbf{x} = -\eta \nabla f(\mathbf{x})$$

$$\boxed{\mathbf{x}_{\text{new}} = \mathbf{x} - \eta \nabla f(\mathbf{x})}$$

Interpretation:

- Gradient = direction of steepest increase
- Negative gradient = direction of steepest decrease

References (Selected)

- Strang, *Linear Algebra and Learning from Data*, 2019.
- Trefethen & Bau, *Numerical Linear Algebra*, 1997.
- Murphy, *Probabilistic Machine Learning*, 2022.
- Bishop, *Pattern Recognition and Machine Learning*, 2006.
- Shalev-Shwartz & Ben-David, *Understanding Machine Learning*, 2014.

Conclusion

Key Message

Linear algebra is the language of data, models, and learning—even when methods become non-Euclidean, the computation rests on $\mathbb{R}^n$ insights.

Thank You!

For your attention and participation !!!

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